

Anthar-Jala Watch – Product Requirements Document (PRD)

Project Type: Android App Development using GenAI

Domain: Natural Resources / Water Sustainability

Target Users: Farmers, Homeowners, Village Communities, Local Water Authorities

1. Problem Statement

Groundwater levels are falling rapidly in many villages and semi-urban areas. Farmers and residents usually realize their borewell is failing only after water availability becomes critically low. There is currently no shared community platform that provides visibility into groundwater health or borewell performance.

2. Vision

Anthar-Jala Watch is a community-powered borewell monitoring application. Users can crowdsource borewell information such as water depth, yield, and drilling year. The system aggregates this data to generate a real-time water stress heatmap for villages and neighborhoods. The platform promotes sustainable groundwater management and encourages recharge activities instead of excessive borewell drilling.

3. Objectives

- Create awareness about groundwater depletion.
- Enable data-driven community water management.
- Predict water stress and borewell failures early.
- Encourage groundwater recharge practices.
- Build a scalable Android platform using GenAI-assisted workflows.

4. Core Features

Borewell Log

Users enter borewell depth, drilling year, current water yield, and optional notes.

Water Stress Heatmap

A Google Maps-based heatmap visualizes groundwater health using Green, Yellow, and Red zones.

Recharge Guide

DIY guidance for recharge pits, rainwater harvesting, and borewell recharge structures.

Alerts & Notifications

Users receive warnings when groundwater levels in their area significantly decline.

5. User Flow

1. User signs in or registers.
2. User enters borewell data.

3. App validates and stores the entry.
4. Heatmap updates dynamically.
5. Users view trends, alerts, and recharge recommendations.

6. Technical Requirements

- Android App developed using Kotlin or Flutter.
- Google Maps API with Heatmap Layer integration.
- Firebase or cloud database for crowdsourced data storage.
- GPS clustering for calculating average groundwater depth.
- Privacy protection by masking exact house locations.
- GenAI integration for smart alerts and recharge recommendations.

7. UI/UX Requirements

- Simple and multilingual interface for rural accessibility.
- Vertical scale visualization for water depth.
- Color-coded maps for easy interpretation.
- Offline-friendly lightweight design.
- Minimal data entry steps for quick usage.

8. Impact Goals

- Promote sustainable groundwater usage.
- Prevent excessive borewell drilling.
- Predict seasonal water stress.
- Increase public awareness about groundwater conservation.

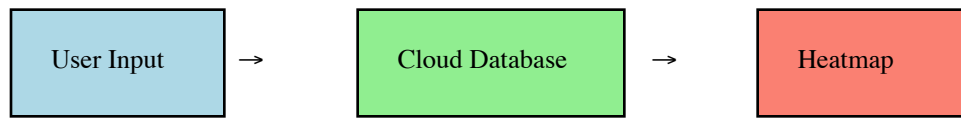
9. Success Criteria

- Heatmap updates accurately based on user inputs.
- Recharge guide includes simple diagrams or sketches.
- Alerts trigger correctly for falling groundwater levels.
- User privacy is protected by hiding precise house numbers.
- Community participation steadily increases.

10. Future Scope

- AI-based groundwater prediction models.
- Integration with government water departments.
- IoT sensor-based automatic borewell monitoring.
- Satellite and rainfall data integration.
- Village-level sustainability analytics dashboard.

11. Concept Diagram



Prepared for: Android App Development using GenAI – Academic/Innovation Project

Project Title :76

Android App Development using GenAI- Anthar-Jala Watch (Natural Resources)

1. The Problem Statement:

Groundwater (Anthar-Jala) levels are falling. Most farmers and residents only realize their borewell is failing when it's too late. There is no "Community Map" of water table health to help people plan their water usage.

2. Detailed Description (The Vision)

Anthar-Jala Watch is a "Community Borewell Monitor." It is a crowdsourced water-table map. Users (farmers/homeowners) log the "Depth of Water" or "Yield" of their borewell. The app aggregates this data to show a "Water Stress Heatmap" for the village. It helps the community decide when to stop digging more borewells and start "Recharge" activities.

3. App Usage & User Flow

- **Borewell Log:** Enter depth, year of digging, and current yield (e.g., 2 inches).
- **Water Map:** A color-coded map (Green/Yellow/Red) of the neighborhood's water health.
- **Recharge Guide:** Simple DIY steps for "Borewell Recharge" structures.
- **Alerts:** "The water table in your area has dropped by 10ft this summer."

4. Technical Implementation & Hints

- **Maps:** Use Heatmap Layer in Google Maps API.
- **Logic:** Calculate "Average Depth" for a specific GPS cluster.
- **UI:** Use a Vertical Scale visual to show water depth levels.

5. Impact Goals

- **Water Sustainability:** Moving from "Individual Extraction" to "Community Management."
- **Disaster Prevention:** Predicting borewell failures before the peak of summer.
- **Resource Awareness:** Making the "Invisible" groundwater visible to the public.

6. Success Criteria for Students

- The map must update its "Heat" intensity based on user inputs.
- The "Recharge Guide" must include simple diagrams/sketches.
- The UI must prioritize "Data Accuracy" and "Privacy" (Hide exact house numbers).